Magnesium Sulphate

Mechanism of action

- Magnesium is the active ingredient in magnesium sulphate.
- Magnesium reduces bronchial smooth muscle contraction resulting in bronchodilation.
- Magnesium is thought to have multiple mechanisms when treating eclampsia, including vasodilation, blood-brain barrier protection, reduction of cerebral oedema, and central anticonvulsant effects.
- Magnesium has been shown to stablise the cardiac membrane during torsades de pointes.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Bronchospasm secondary to severe or immediately life-threatening asthma.
- ✓ Bronchospasm secondary to severe COPD or imminent respiratory arrest from COPD.
- ✓ If torsades de pointes is present or has occurred.
- ✓ Suspected eclampsia.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Hypotension. Magnesium is a vasodilator and may make hypotension worse.

Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.

Magnesium Sulphate

Dosage

- Asthma:
  - 10 mmol (2.5 g) IV for an adult.
  - See the paediatric drug dose tables for a child.
  - A second dose may be administered if transport time is longer than 30 minutes and the patient is not improving.
- COPD:
  - 10 mmol (2.5 g) IV for an adult.
  - A second dose may be administered if transport time is longer than 30 minutes and the patient is not improving.
- Torsades de pointes is present or has occurred:
  - 10 mmol (2.5 g) IV for an adult.
  - Repeat once if episodes of torsades de pointes continue.
- Suspected eclampsia:
  - 20 mmol (5 g) IV.

Administration

- Cardiac arrest if torsades de pointes is present or has occurred:
  - Dilute to a total of 10 ml using 0.9% sodium chloride and administer over 1-2 minutes.
- All other indications:
  - Administer IV over approximately 15 minutes.
  - Draw up the dose and dilute to a total of 10 ml using 0.9% sodium chloride (if required) and administer 1 ml every 1-2 minutes, preferably into a running line.
  - Alternatively, in adults or children weighing over 40 kg only, add to a 100 ml bag of 5% glucose, shake well, label and infuse over approximately 15 minutes:
    - 2-3 drops per second via a standard IV administration set will deliver 100 ml over approximately 15 minutes.
    - The administration set will need to be flushed with 0.9% sodium chloride to ensure all the magnesium has been administered.

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Common adverse effects

- Flushing. Particularly if administered rapidly.
- Hypotension. Particularly if administered rapidly.
- Muscle weakness. This is usually only seen with doses exceeding 20 mmol.

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 30-60 minutes.

Pharmacokinetics

- Magnesium is primarily excreted in the urine.
- There are no significant effects from renal impairment on the initial (loading) dose.

Usual preparation

- 5 ml ampoule containing 10 mmol (2.5 g) of magnesium.

Common interactions

- None.

Additional information

- Do not administer magnesium into an IV line that has an adrenaline infusion running through it concurrently, because precipitation may occur.
- Rarely, magnesium may have a role in the treatment of severe pre-eclampsia. In this setting seek clinical advice.
- Document the dose administered in mmol.

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More Information

Version history

9 June 2025 1.0.5.3

Update: Dosage updated.

11 September 2023 1.0.4

Update: Indications have been added. Dosage, administration and additional information has been updated.

22 November 2022 1.0.1

Update: Indication for COPD.